

Audio Tactile Map for the Gannett Art Gallery Final Report



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Introduction

The Creating Collaboratively Across Visual Abilities (CCAVA) project, funded by a SUNY Innovative Instructional Grant, focuses on improving accessibility for visual-oriented research and creative exhibitions and overcoming barriers related to capacity and economic status. In today's digital age, 3D mapping offers dynamic ways of visualizing spatial data, making it more accessible to people with low vision or blindness, with applications ranging from games to urban planning. During the spring semester of 2024, we designed and 3D-printed the first prototype of an audio-tactile map for the Gannett Gallery, aimed at improving exhibition access for people with visual disabilities. Through the Capstone project, there is a desire to design an improvement for the audio-tactile map. In order to develop the next version of the map, we will utilize the knowledge we've learned from the past years at SUNY Polytechnic such as Computer Aided Design, Finite Element Analysis, Electrical Circuits and Assistive Technology. We aim to achieve a modular map that can be used for multiple exhibitions in our Gannett Gallery.

Equipment

- SolidWorks: This our primary CAD software that we used to re-design and improve the audio tactile map
- ANSYS: The software provides the tool to model the ATM map using the finite element method for stress analysis and understand the potential failure points for design or material properties.
- Creality Ender 3 S1 Pro 3D printer: Provided by SUNY Polytechnic Institute's CGAM. These printers will be our primary machine in creating the map.
- Arduino MEGA: This microcontroller will be the brain of the electrical system inside the map.
- Speaker and headphones: these are the two audio output options that users can choose based on their preference.
- Micro SD Card Reader: this device stores all the audio files of the artworks for the Arduino to locate and use.

- Buttons: our main source of input to play the audio.
- Polymaker Polysmooth PLA: this material is the most suitable for the 3D map as it will give the best smoothness, ensure comfortability for users.
- PLA: this material is the most common for 3D printers.
- Breadboards: Used for electronic systems testing.
- Solderable breadboard: Used for connecting the ground legs of all buttons.
- Switch 3 Pin 3 Position: a switch to choose between headphone and speaker.
- Stereo Audio Balanced Male & Female Jack to AV: An adapter to connect headphones to arduino.

Table 1: *Supplies and Costs of Necessary Parts*

Already Obtained	
Parts	Costs
Arduino MEGA	\$48.40
Speaker	\$10.19 (Packs of 4)
Headphones	Provided by Gannett Gallery
Micro SD Card Reader	\$6.99 (Packs of 5)
MatterHackers PLA	\$20.89 (per roll of 1kg)
Breadboard	Provided by SUNY Poly
Solderable breadboard	\$7.99 (Pack of 5 + 1 mini board)
Need to be purchased	
Polymaker Polysmooth PLA	\$36.99 (per roll of 0.75kg)
Switch 3 Pin 3 Position	\$6.99 (pack of 5)
Stereo Audio Balanced Male & Female Jack to AV	\$11.73 (pack of 8)

Theory

The project was mainly done by utilizing CAD (Computer Aided Design). It is the use of computer-based software to aid in the design processes. CAD's purpose is to optimize and streamline the designer's workflow, increase productivity, improve the quality and level of detail in the design, improve documentation communications and often contribute toward a manufacturing design database. The CAD software outputs are usually in electronic files and are used accordingly for manufacturing processes. In our case, SolidWorks is our primary CAD software.

In mechanical design, stress analysis is crucial to determine the material properties for design, this procedure helps a designer to predict how a structure behaves under different mechanical conditions. This analysis is also an assessment of the factor safety to ensure the design fulfills the standard requirement.

The finite element method is a theoretical approach for the analysis of linear elastic materials which offers insights into how solid structures deform when subjected to applied loads. Still, since PLA plastic is a thermoplastic polyester material and inhomogeneous, its properties would conflict in 3D printing due to cross-section differences and interfacial areas between layers. Such interfaces may lead to the failure of the object. In addition, it suffers from a layer-by-layer manufacturing process, which is complex to define its behavior. For instance, infill density and print direction have a strong effect on the material's strength but also produce a difference in FEA analysis results.

To do a simulation for PLA plastic material on FEA, we made an assumption in our capstone project for an audio-tactile map, we chose PLA plastic material for design. For the PLA material, we made an assumption based on the equation and used the finite element method to verify its stress analysis. The stress analysis is for the bottom layer of the map which is a flat square plate subjected to uniform loading. The assumption for the flat plate is thin with four simple supports at the bottom, it was made of an elastic, homogeneous, isotropic material.

The following equations are used for stress analysis:

$$P = \frac{F}{A}$$

While:

F: Distributed Loads (lb)

A: Plate Area (in^2)

Maximum deflection at center,

$$y_{max} = \frac{0.142pa^4}{Et^3[2.21(\frac{a}{b})^3+1]}$$

Bending stress at plate center,

$$\sigma = \frac{0.75pa^2}{t^2[1.61(\frac{a}{b})^3+1]}$$

While:

y_{max} : Maximum deflection at center of plate

σ : Bending stress at center of plate

p: Uniform pressure at the top of plate

a: Length of the plate

b: Width of the plate

t: Thickness of the plate

E: Young's Modulus Elasticity

Procedure

As mentioned in the introduction section of this document, there was a previous prototype that was developed in the semester of Spring 2024. The future builds or versions of the ATM will be largely based on that prototype. The new and improved version improves on many weaknesses of the prototype.



Figure 1: *Original Prototype*

Before immediately developing the new version of the ATM, there were parameters that needed to be met before the new version was made.

Parameters required to meet,

- Larger frame
- Modular design
- Implementation of ports
- Access to battery

By examining and implementing these new parameters into the new design, SolidWorks was chosen to be the 3D Computer Aided Design software due to the amount of experience and the ease of use when designing 3D artifacts. However, since the new design is going to be based on the original prototype, there will be quality of life improvements for both the end user and the person who is going to be maintaining the device. Our improved design will keep the initial design's framework but add on and improve certain aspects.

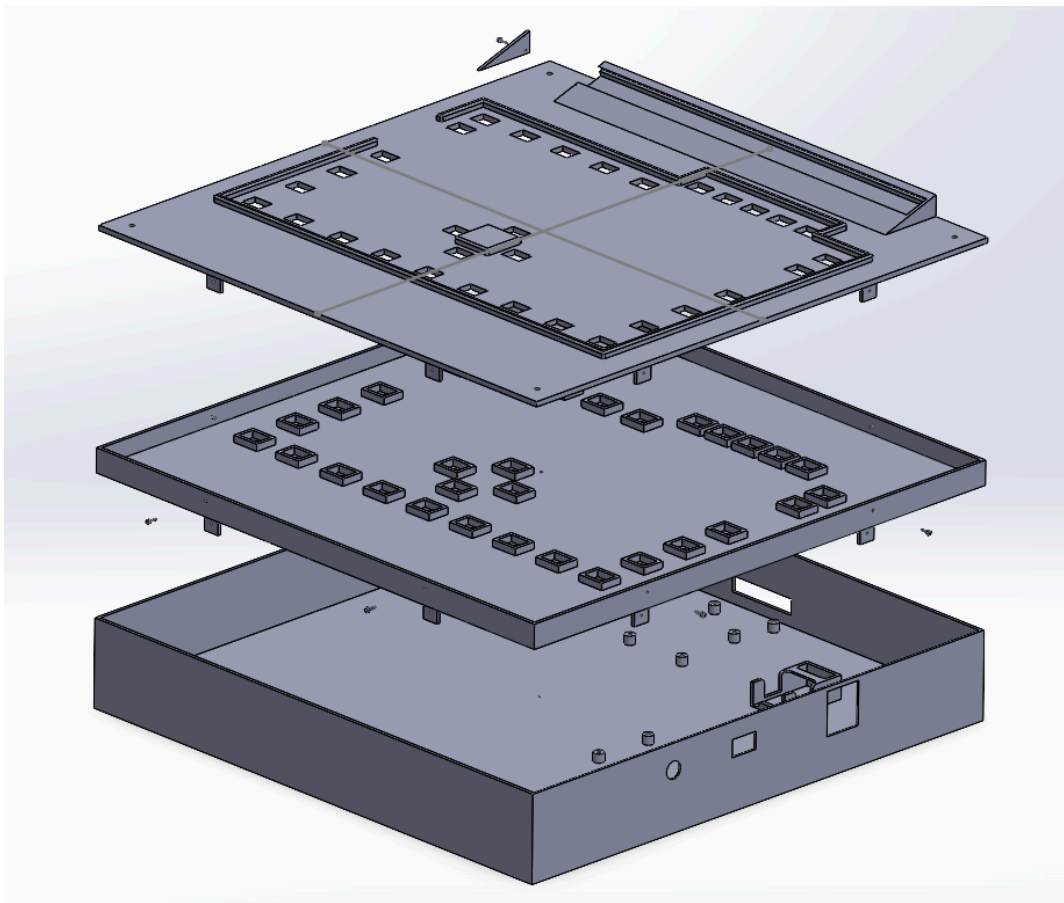
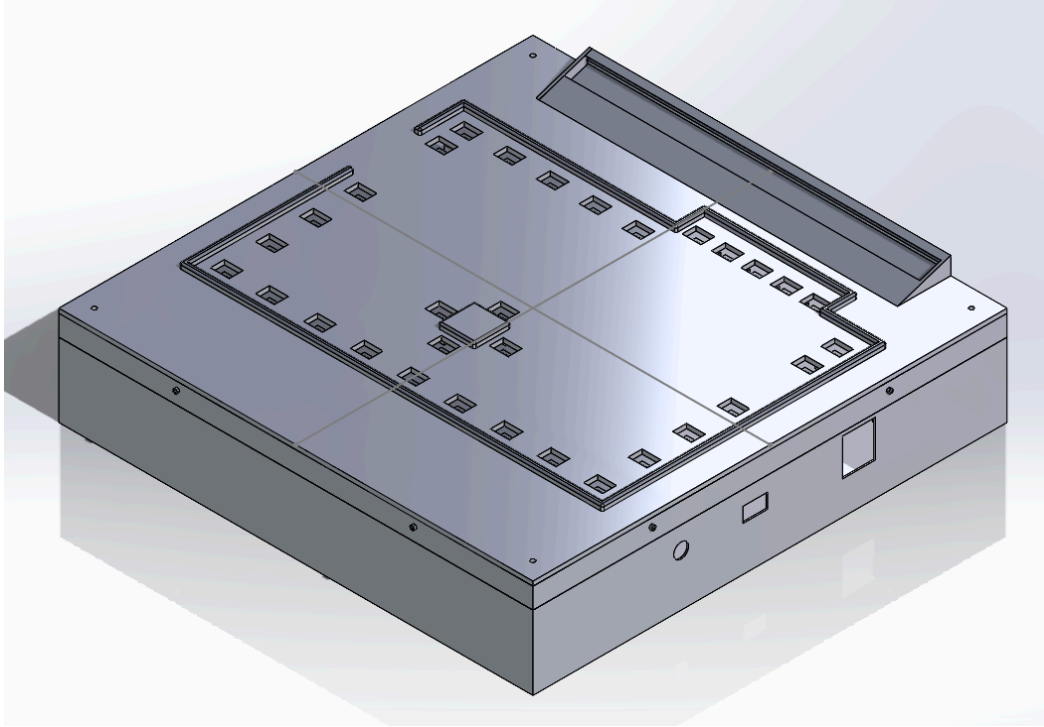


Figure 2: Assembly and exploded view of new Audio Tactile Map

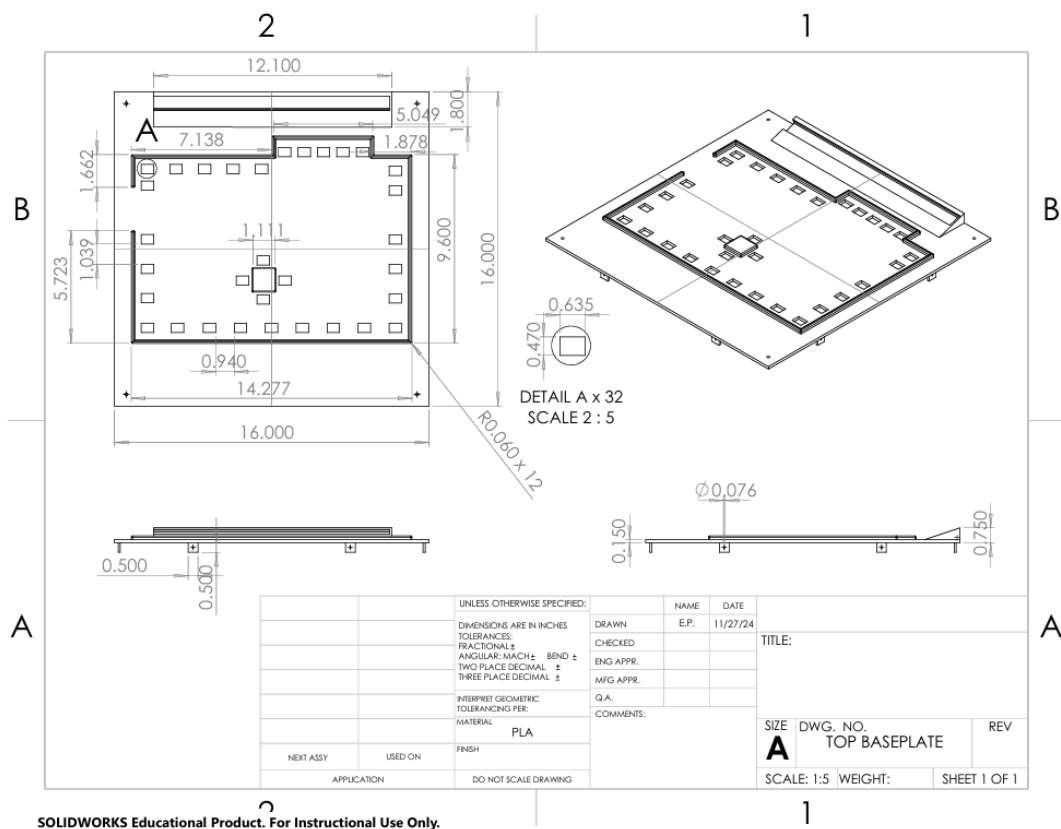
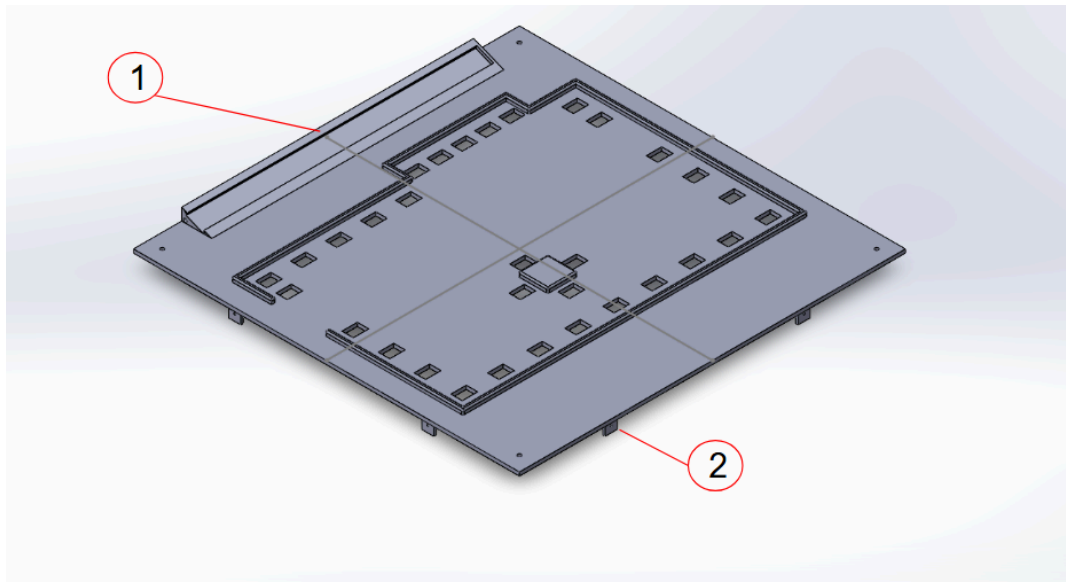
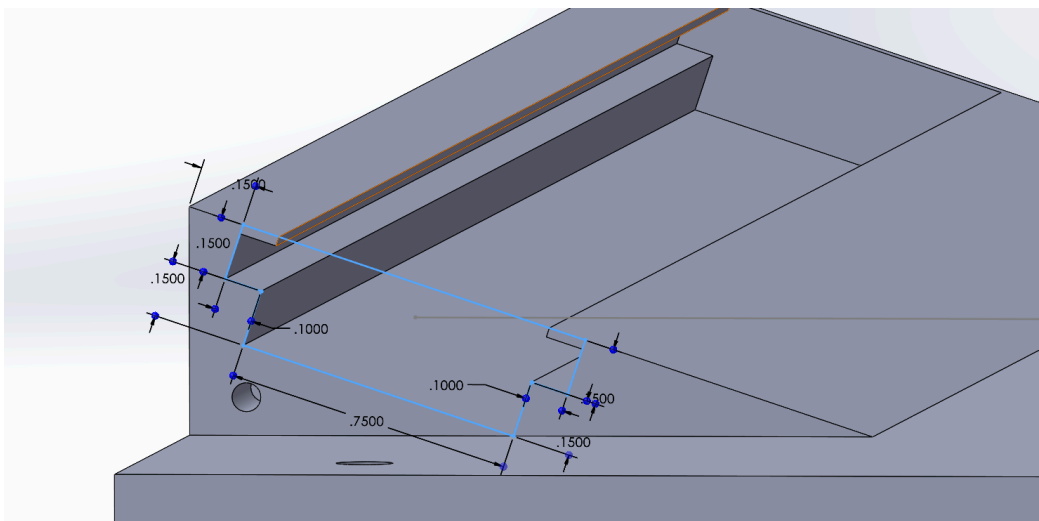
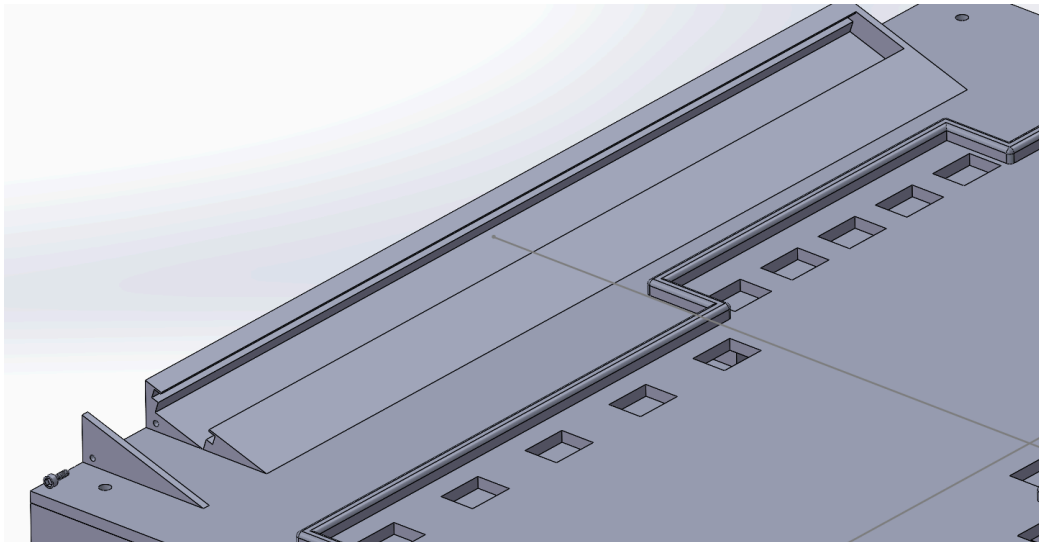


Figure 3: Top Plate

Figure 3 represents the updated top plate of the artifact where it has rectangular holes that will fit braille blocks that signify the different art pieces that will be displayed in

the art gallery. The contours in Figure 3 is the general outline of the art gallery. In total, there are 32 buttons that may be utilized for providing information to the user. The intention for the holes is the capability of being modulated so that when there is not an art piece available in that slot of the map, then the user is not able to interact with the button, therefore having the user move onto the other buttons using the sense of feeling and contact.



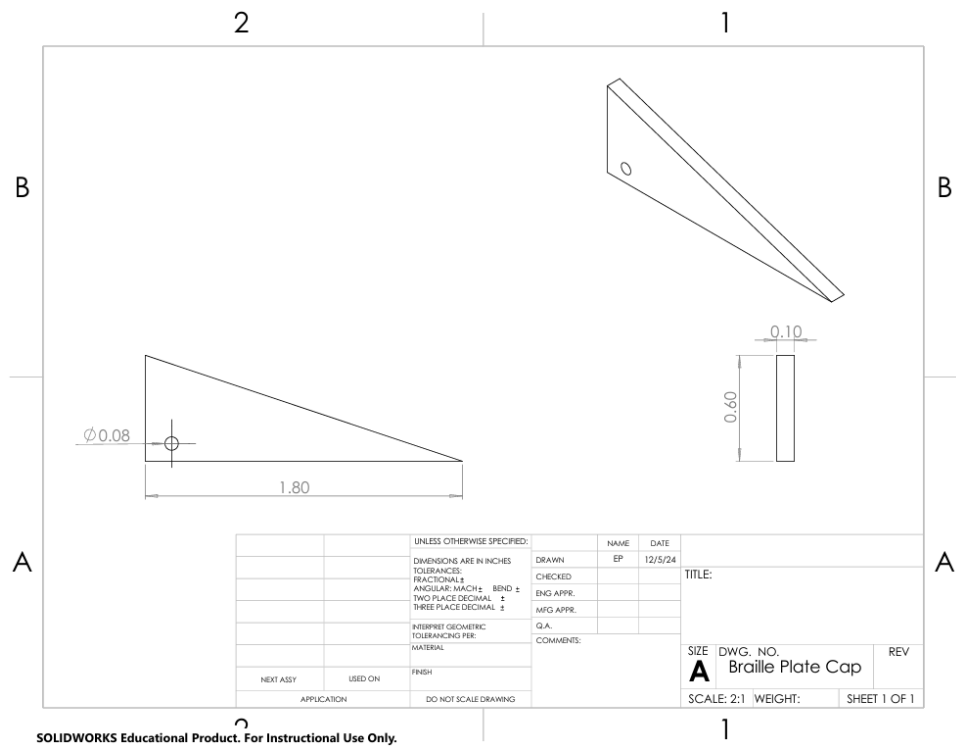
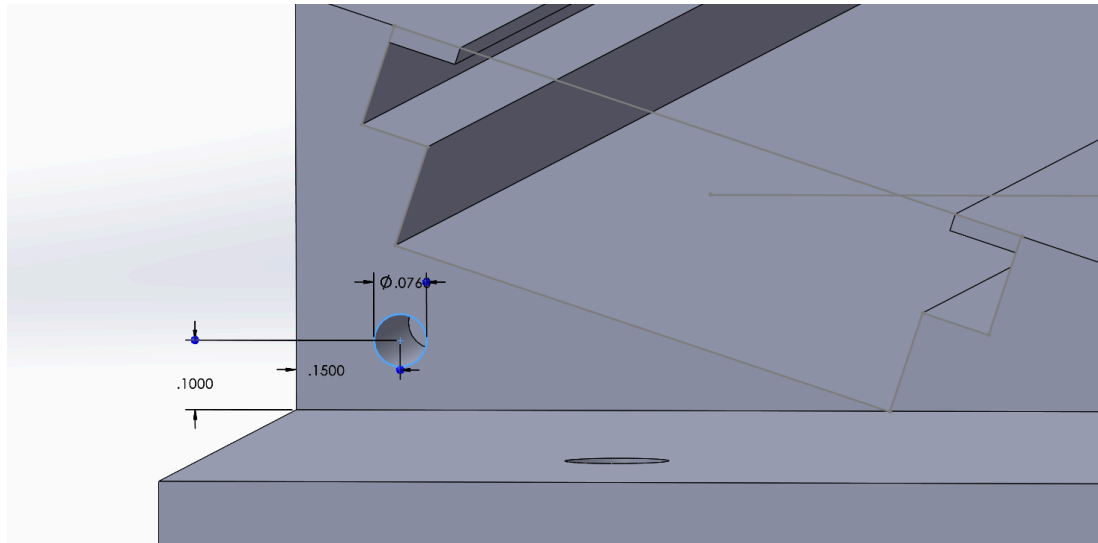


Figure 3 Section 1: Braille Plate Insert

In section 1 of Figure 3, that part of the design is a new addition to the design where there is a slot to insert a braille plate from the left side of the insert opening. The plate that is to be inserted into the display can easily be used for different functions. It may be used to give the user the title of the current name of the display or give more

details. Once the plate is inserted into the slot, it will be locked in place using a screw and a cap to seal off the slot.

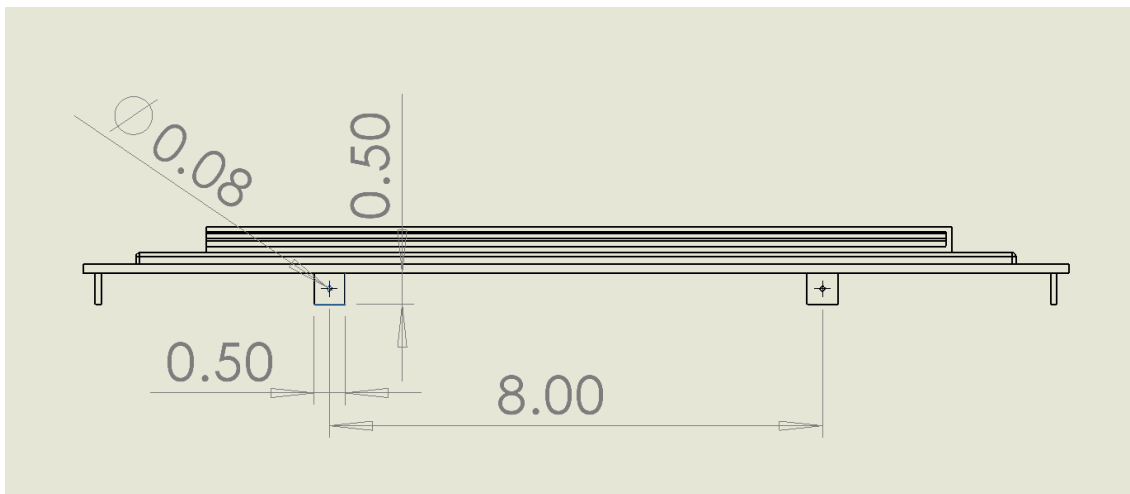
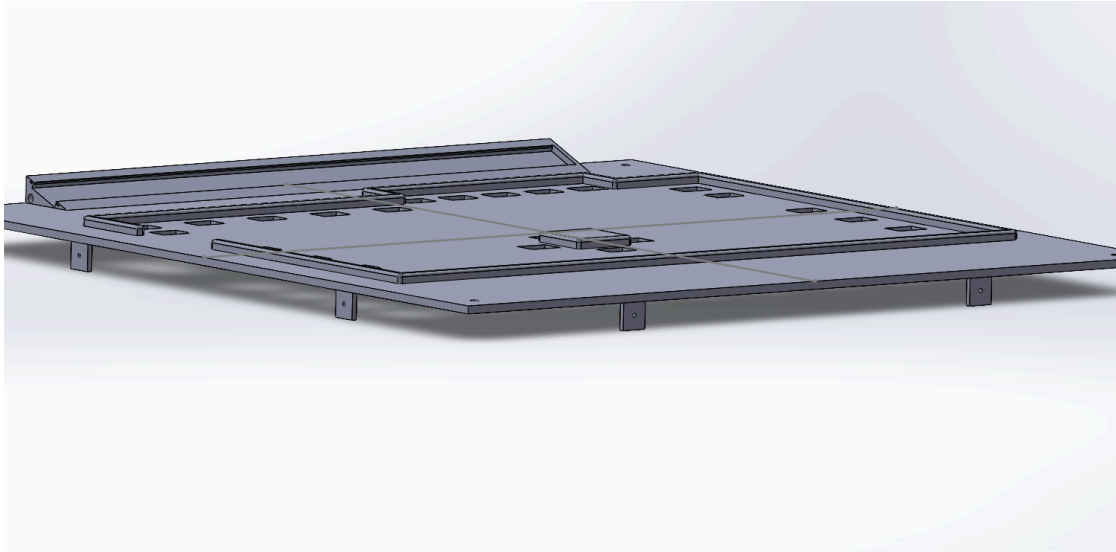


Figure 3 Section 2: Screw Tabs

Displayed in Figure 3 Section 2, these are the screw tabs in order to keep the ATM together. These holes will be part of the 3D printed design and the holes should fit a standard M3 screw referenced in Figure Section 3. These tabs are present in the top

baseplate and the second layer of the ATM. These screw tabs are located exactly the same as the other layers of the ATM.

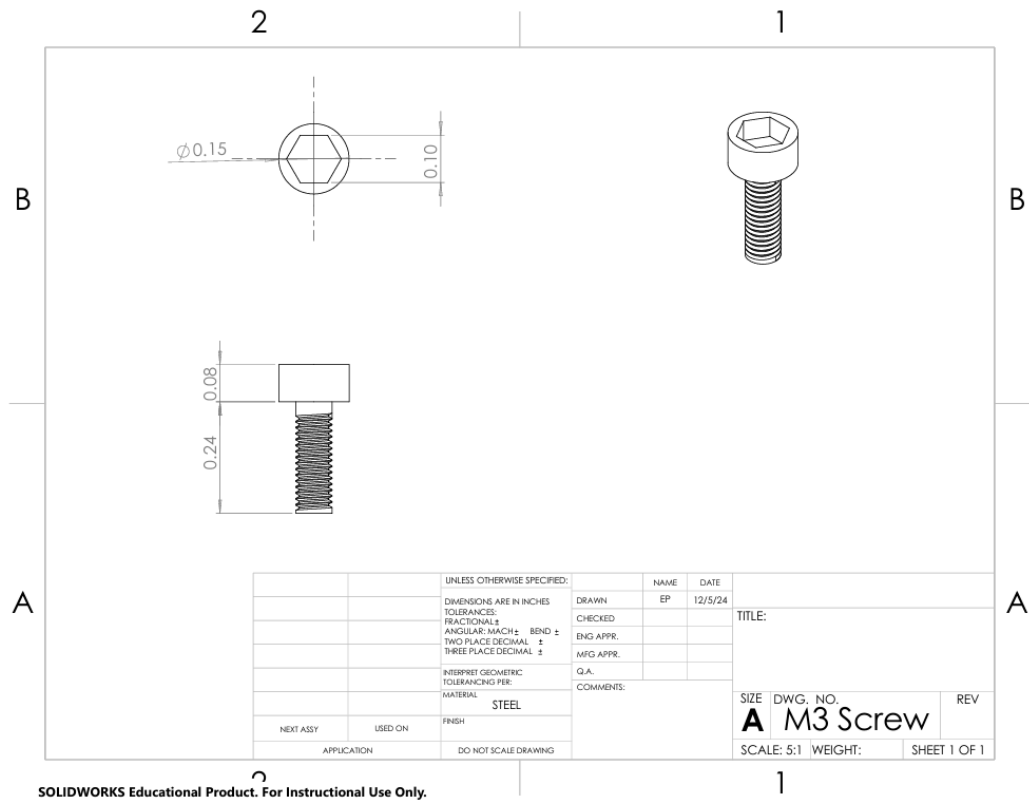


Figure 3 Section 3: M3 Screw

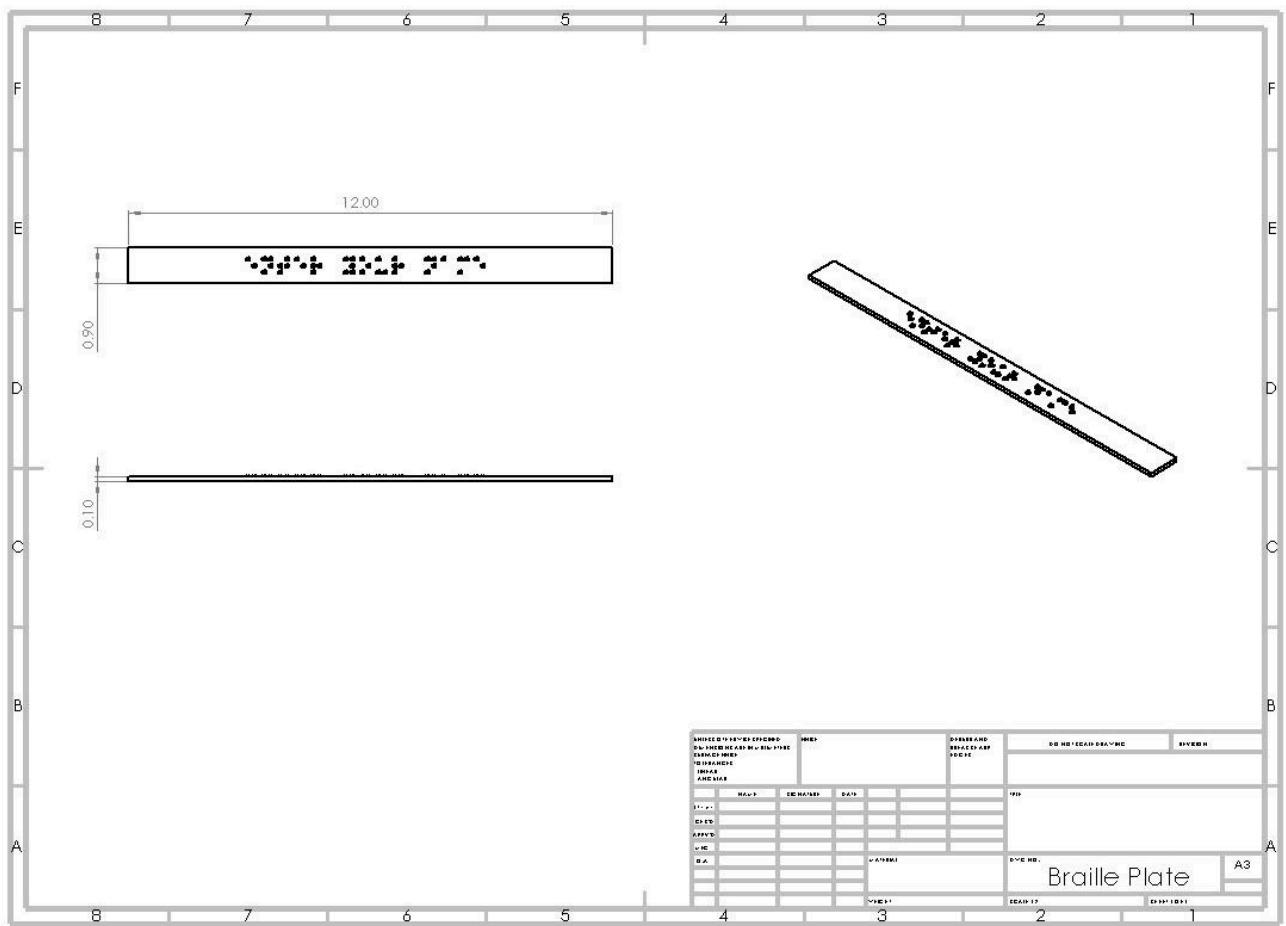
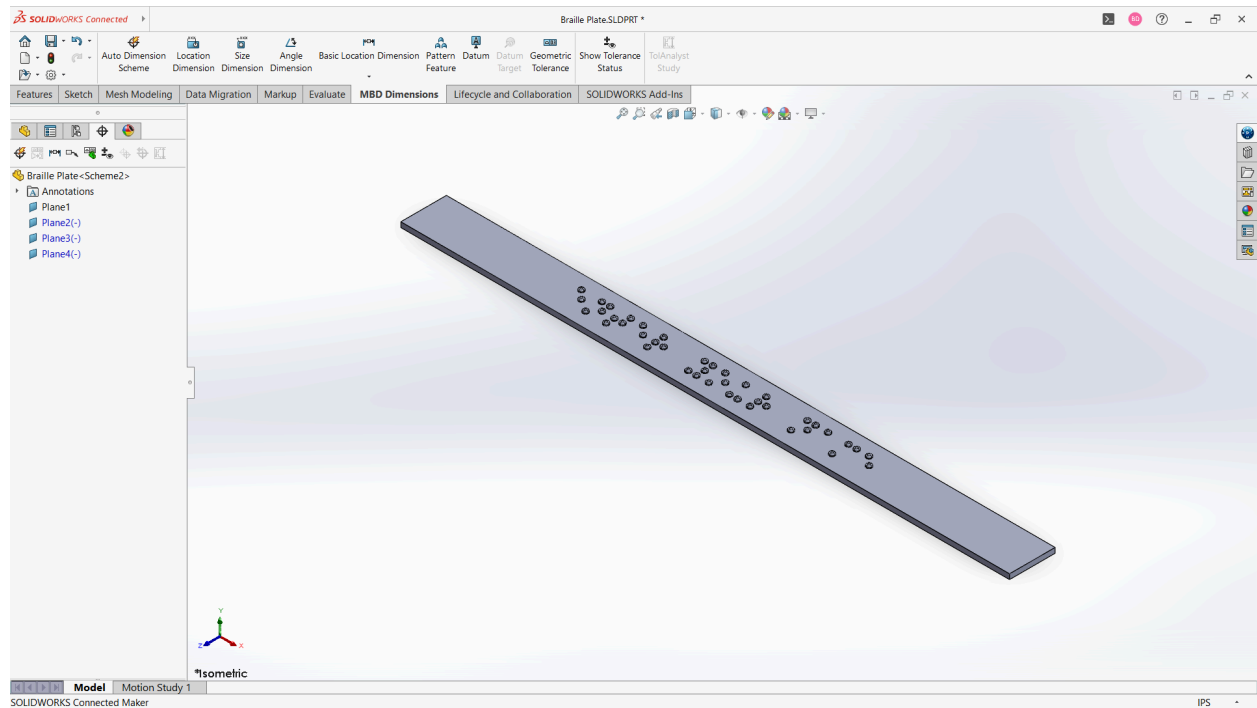
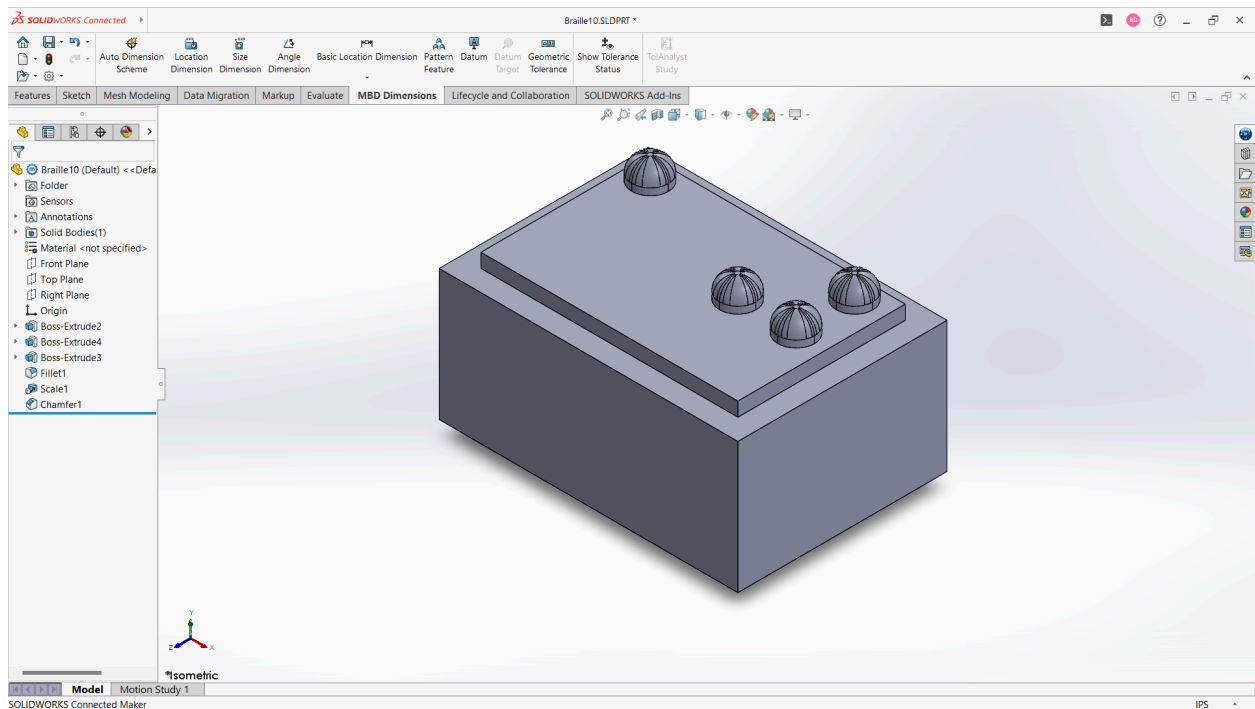
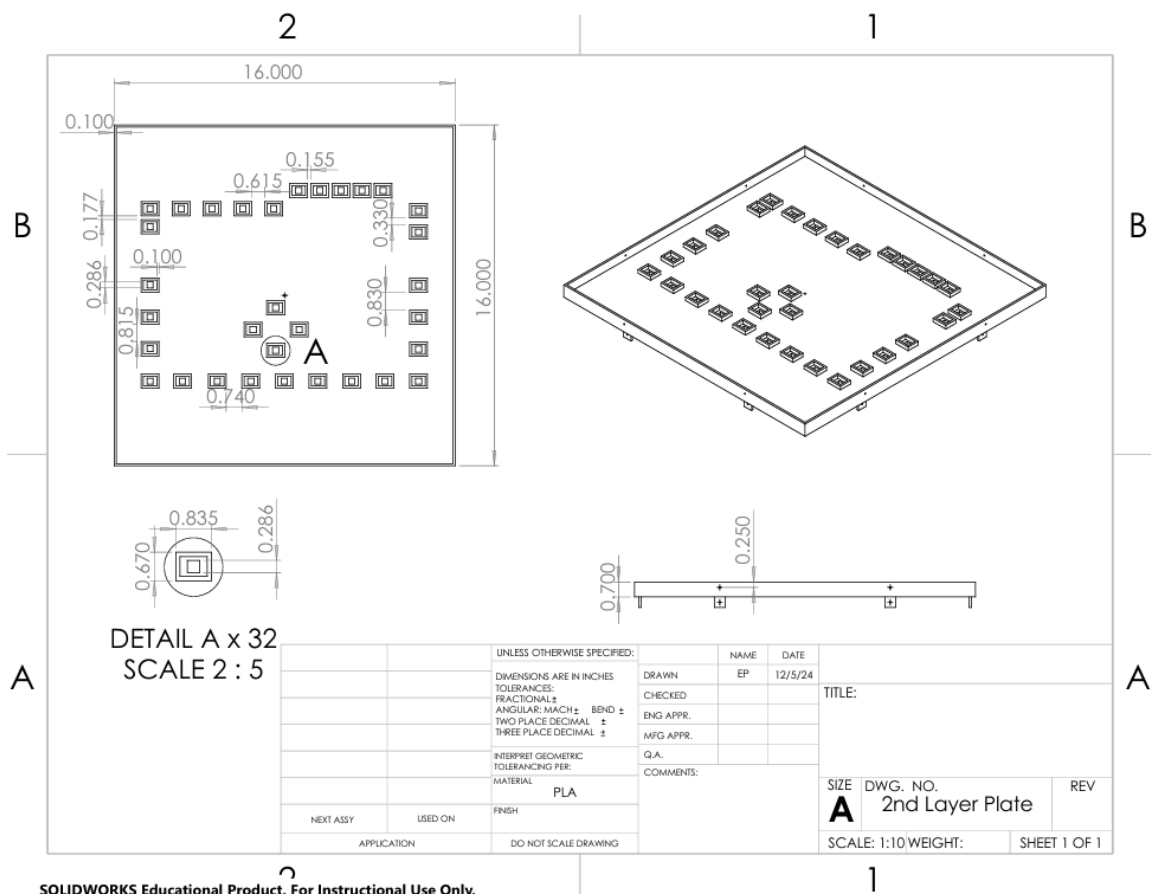
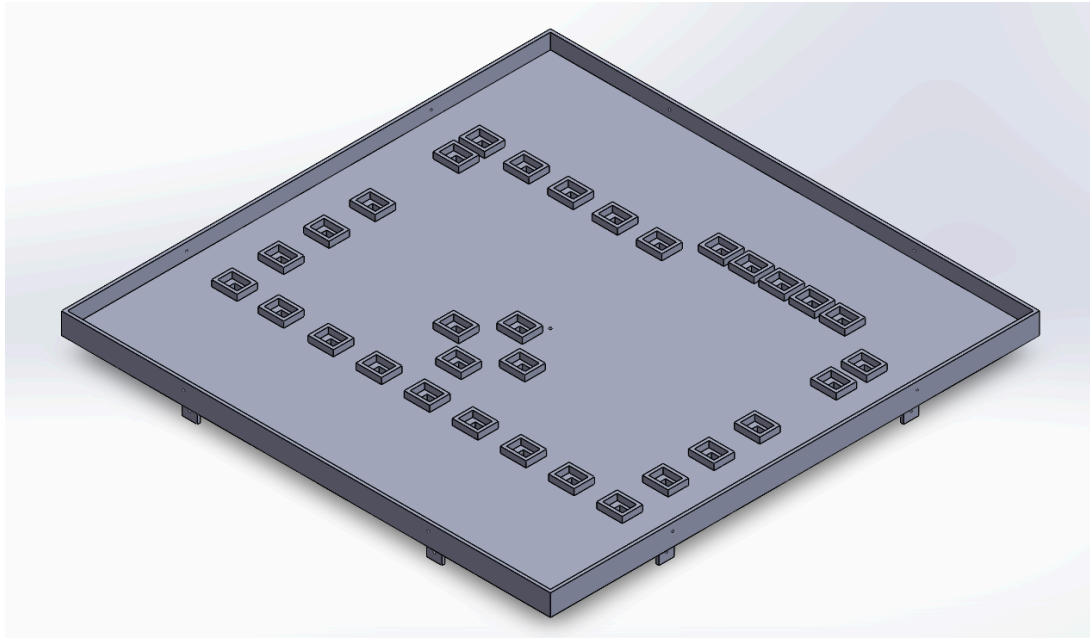


Figure 4: *Braille Plate*

Braille blocks will also be integrated into the first layer of the map to indicate the locations of artworks. These blocks, numbered 1 to 32, correspond to the artworks on display in the exhibition and are arranged in a clockwise direction, starting from the left side of the map. When an artwork is not being displayed at a specific location, the corresponding braille block can be removed and replaced with a blank block.

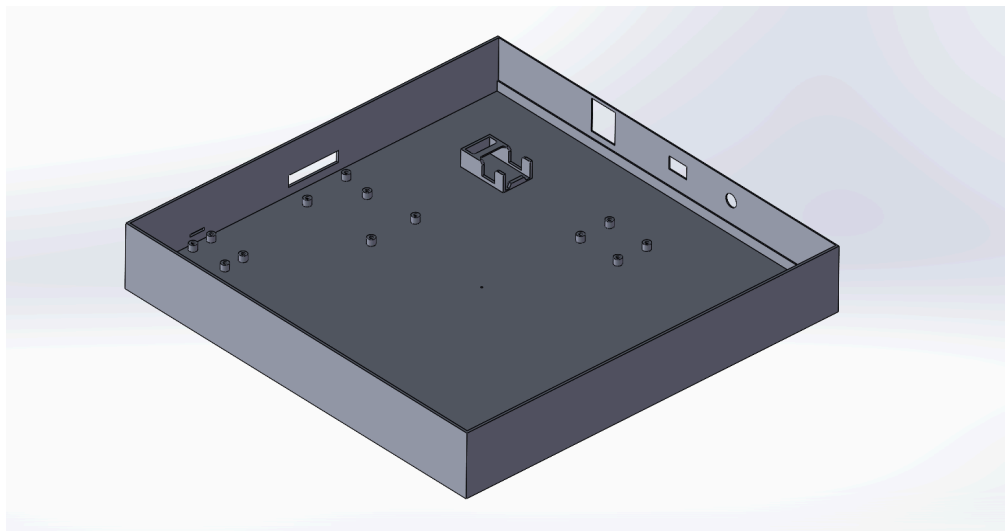




SOLIDWORKS Educational Product. For Instructional Use Only.

Figure 6: *Second Base Plate*

Displayed in Figure 6, the second base plate is a new addition to the design compared to the prototype where it only had the bottom base layer and the upper surface face containing the actual physical buttons that are wired to the electronic components in the bottom layer of the ATM. This second base plate was added to improve the modularity of the overall design of the ATM as well as considerations of the newly implemented Braille Blocks featured in Figure 5. The wired buttons are essentially going to be seated in each hole while the extruded parts of each hole serve as seats or to secure the Braille Blocks when they are inserted to be used. The “walls” of the second base plate are going to be exposed on the exterior of the model which are then secured with M3 screws attaching it to both the top base plate and the bottom layer’s walls as well, ensuring a secure fit with clearance.



jack, and a speaker to convey sound to the outside. All designs and dimensions were shown in detail in the picture and drawing.

A simplified circuit setup is drawn below:

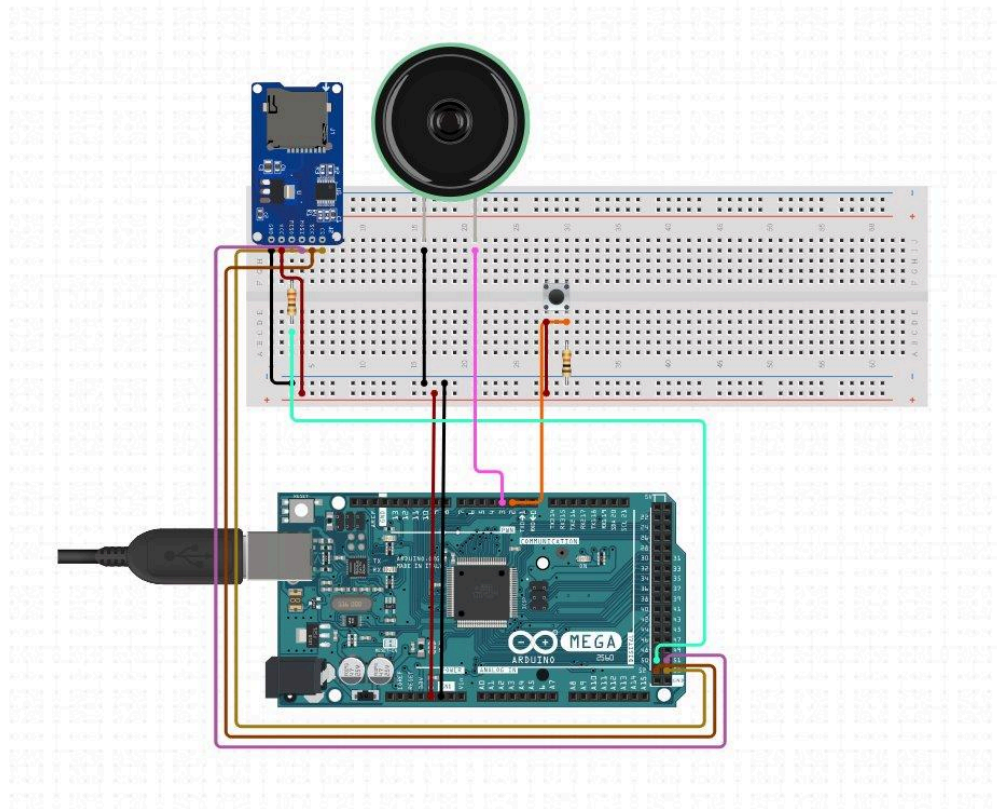


Figure 7: Setup of the electrical system.

The system will have buttons, serve as the input and for the output, users will have the ability to choose between speaker or headphone based on their preference. The microSD card stores all the audio description of every artworks that will be displayed in the exhibition.

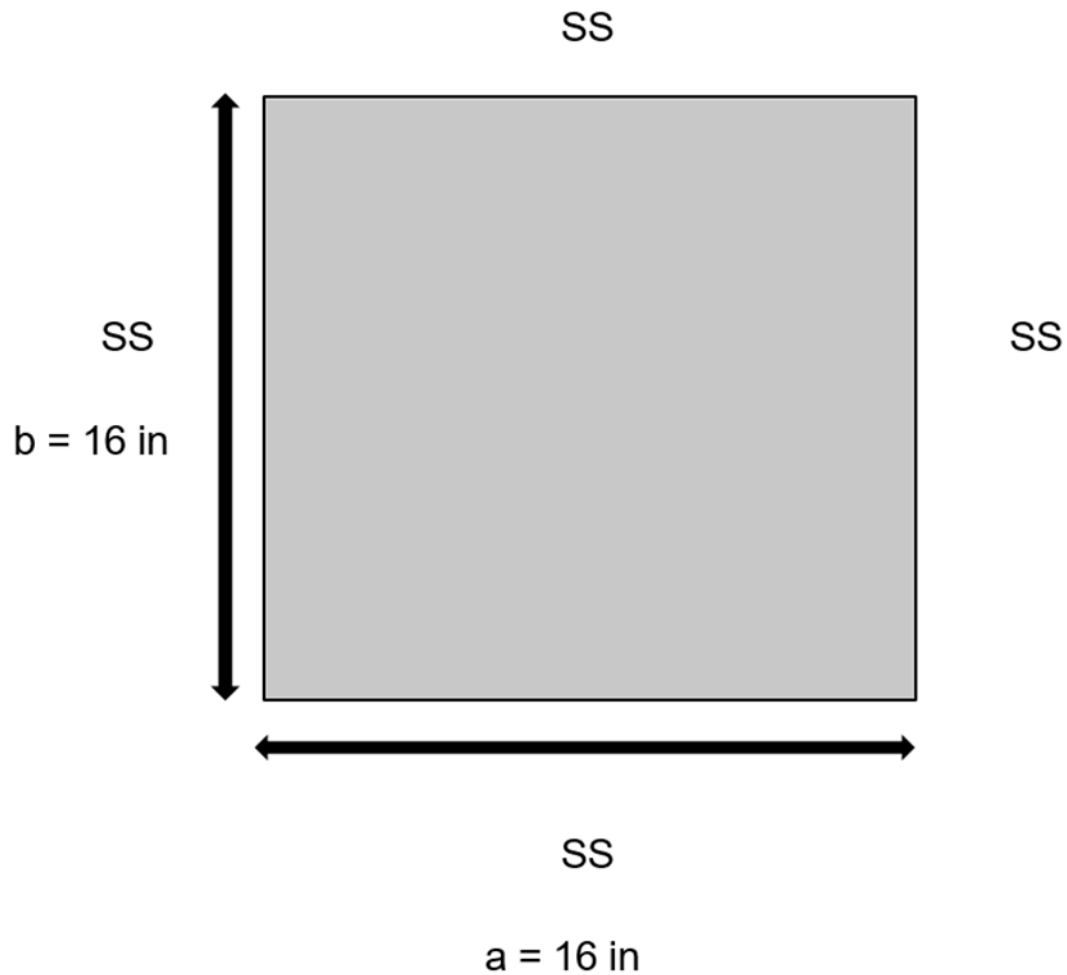
Analysis and Result

We plan to build the ATM with the total weight for the map estimated at 7 oz about 0.4375 lb distributed loads including all electrical component devices, cables, wires, and motherboards.

The material we plan to design and choose will be PLA (Polylactic Acid) plastic, with a thickness of 0.5 in, and a square plate with dimensions W=16 in. With a total weight of 0.4375 lb and an area of a square is 16inx16in = 256 (in^2).

Then we have

$$P = \frac{F}{A} = \frac{0.4375 \text{ lb}}{256 \text{ in}^2} = 1.71 \times 10^{-3} \text{ psi}$$



While:

SS: Simple Support

Young's Modulus Elasticity: $E = 340 \times 10^5 \text{ psi}$

Thickness: $t = 0.5 \text{ in}$

Load: $p = 1.71 \times 10^{-3} \text{ psi}$

Area: $A = 256 \text{ in}^2$

The properties of PLA material required to be used to calculate the stress on Ansys APDL and Workbench includes: Density= 0.047 lb/in^3 , Young's Modulus= 340 ksi, Poisson's Ratio=0.3, Tensile Yield Strength= 6.6 ksi, Tensile Ultimate Strength= 8.31 ksi.

Model the bottom layer plate in 2D by using finite element ANSYS-APDL software. A cutting method was applied to cut the whole plate into 4 elements, each element corresponding to a quarter, and then analysis by modeling a quarter plate element with 2 simple supports and 2 symmetric boundary condition lines.

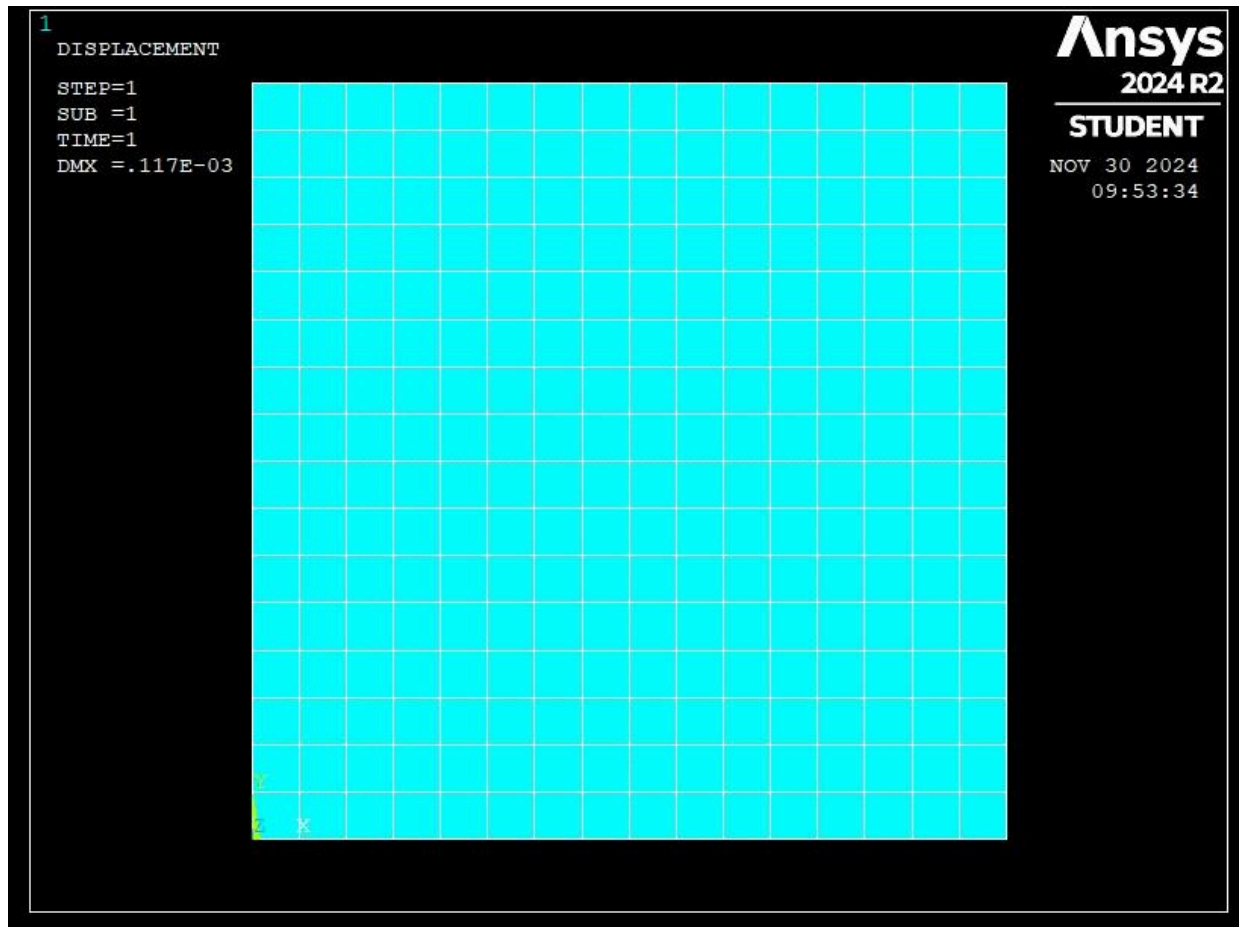


Figure 8: *Maximum deflection at the center of the plate*

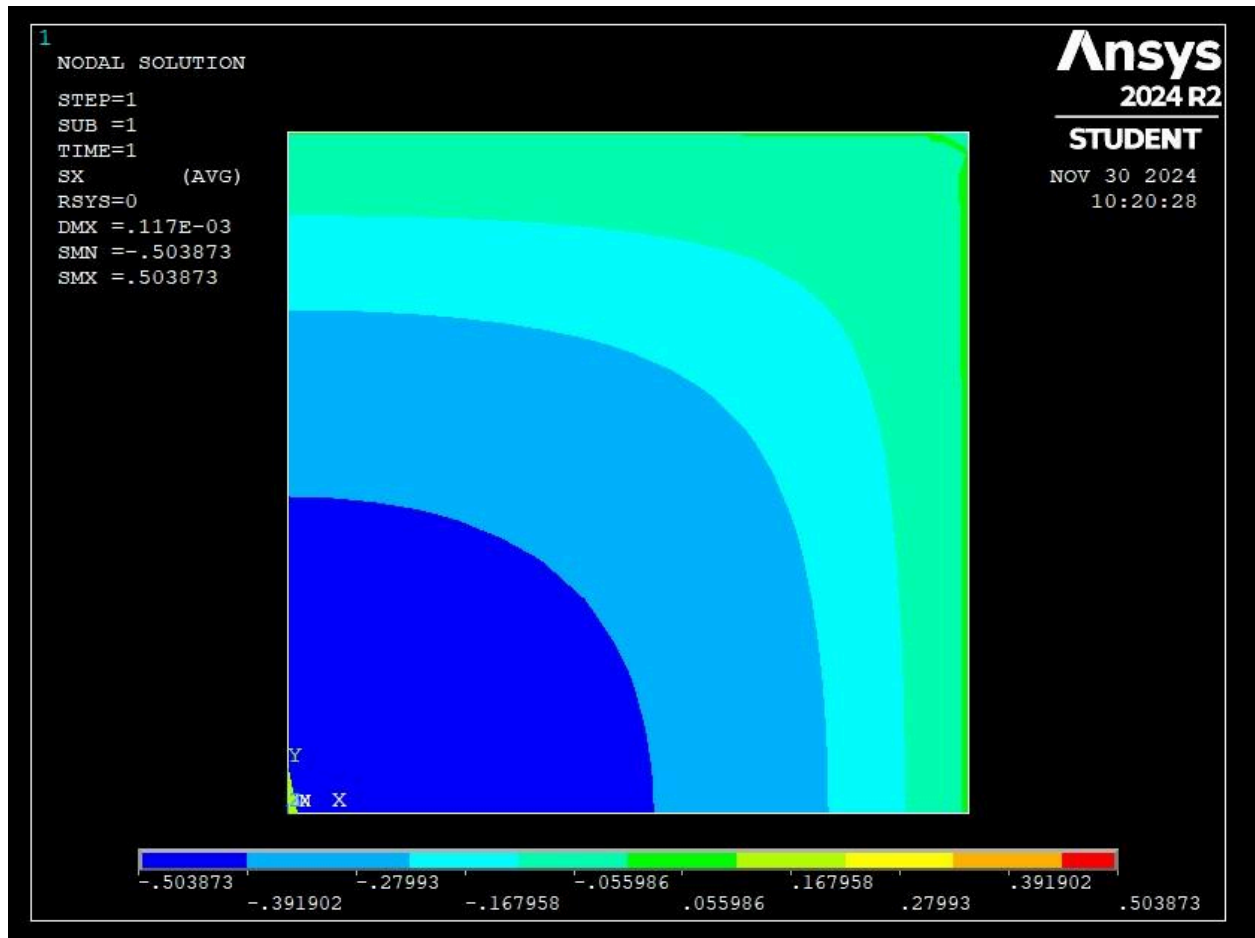


Figure 9: *Bending stress at the plate center*

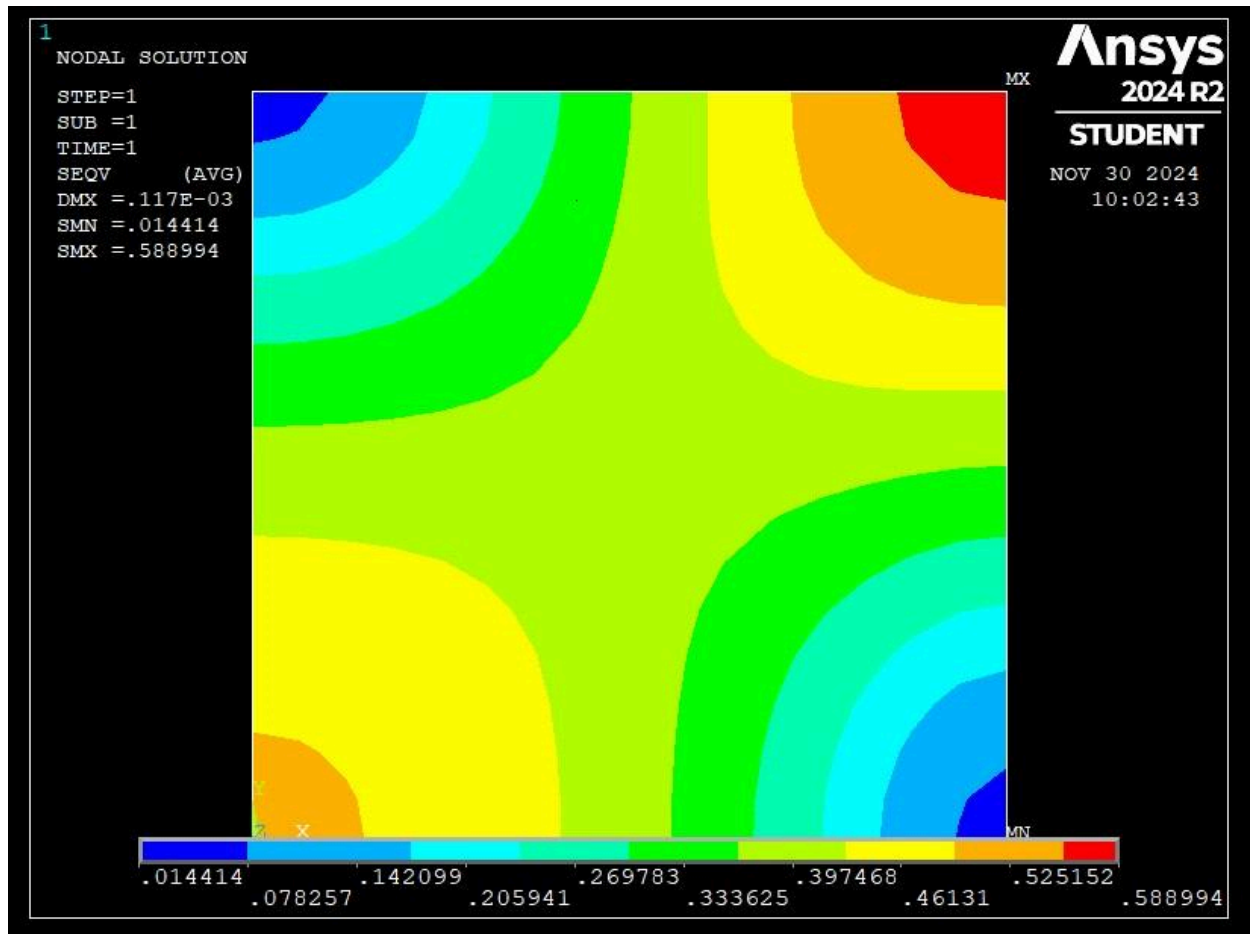


Figure 10: *Von Mises Stress*

The Von Mises stress predicts the yielding of materials under loading based on the tensile test, in this case. The Von Mises Stress = 0.589 psi

When applying loads to the entire ATM map, we examine the properties of the Polylactic Acid (PLA) material and conduct a mechanical stress analysis on it. This analysis aims to verify that a structure will perform its accurate function in a particular load environment. It is critical to anticipate all failure modes and design to fulfill the standard requirement.

Based on the SolidWorks model, we used Ansys Workbench software. We did several steps such as inputting PLA data, importing mesh geometry, and combining boundary conditions, including simple supports and pressure, for the part. We got the

maximum Von Mises stress and safety factor for the bottom layer. By verifying with hand calculations, the result is close to analysis, and the part is safe to design.

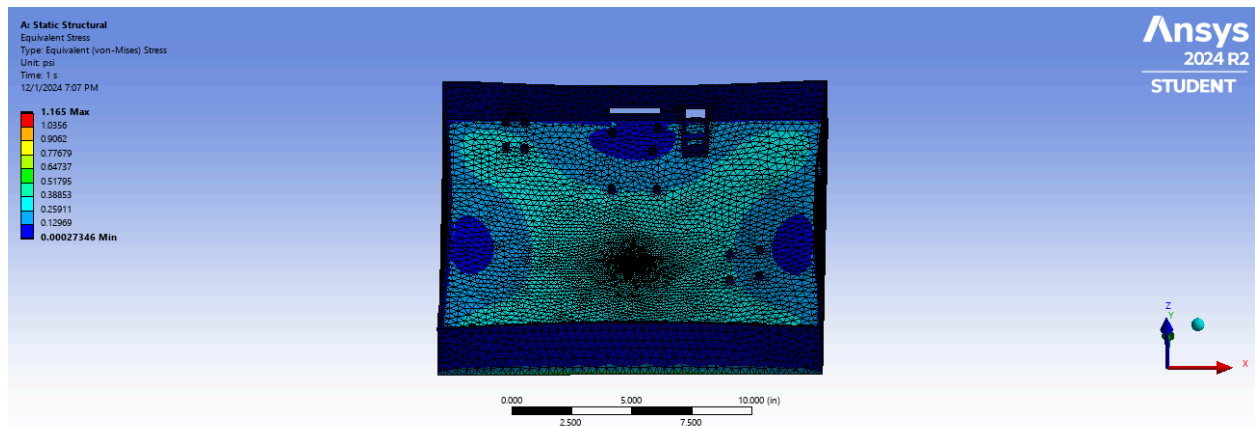


Figure 11: *The Von Mises Stress*

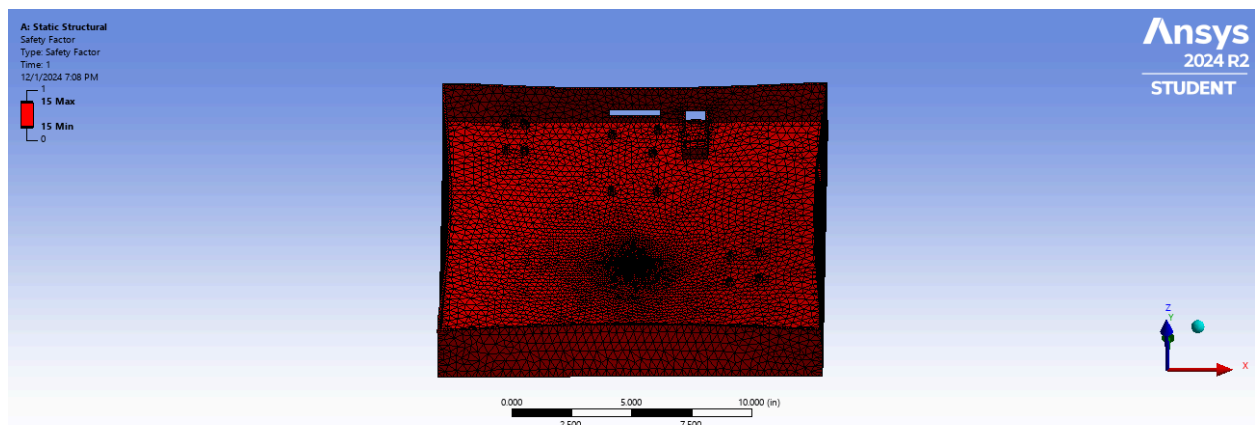


Figure 12: *The factor of safety*

The Von Mises stress is usually used for ductile materials such as metals. Our design uses PLA plastic, which has properties such as being brittle and having less than 10% elongation at break. The figure shows the Von Mises stress result to compare with the normal stress result for prediction purposes.

Discussion

The CAD design of the 3D audio map was successfully completed using SolidWorks, with all necessary parameters and features carefully considered during the design process. The software provided a detailed and accurate representation of the planned design, ensuring that the key components were accurately modelled to meet the project specifications.

However, despite careful design, challenges may arise in the next stage of the project, particularly in achieving accurate dimensions due to the inherent limitations of 3D printing technology. Although appropriate parameters have been taken into account, the precision of consumer-grade 3D printers may vary, resulting in dimensional differences between the design and printed components. Dimensional inaccuracies during the printing process may lead to several challenges, including electronics that cannot be integrated into the map as intended and require post-process adjustments. This reprinting not only delays the project timeline but also increases material usage, resulting in higher filament consumption.

Teamwork

A milestone chart was laid out and completely achieved throughout the project.

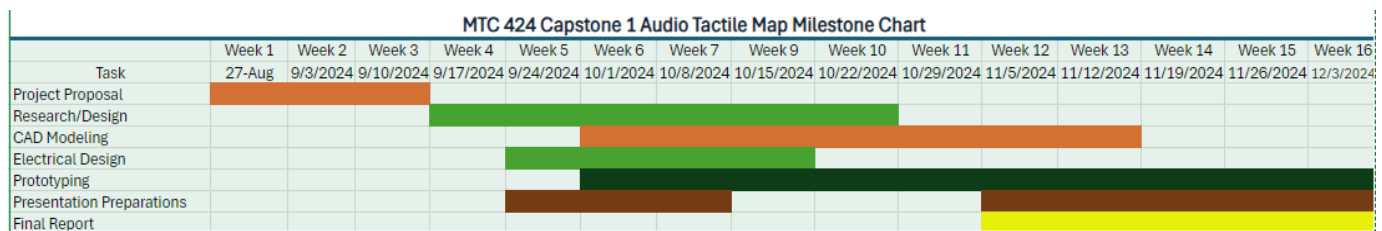


Figure 13: *Project Milestone Chart*

The amount of work was divided evenly within the group:

- Bao Do: focused on the electrical aspect of the map, including electronics wiring and programming.
- Edric Pham: focused on designing the top and middle layers of the audio map

- Cong Phan: focused on designing the bottom layer and stress analysis for the whole map.

Reflection

This project aimed to upgrade and improve the prototype created in the spring semester of 2024. Our main focus was to address the limitations and issues of the original map design to make it more flexible and user-friendly. The goal was to create a modular and scalable map that could fit different exhibitions and artwork placements.

One of the biggest improvements was doubling the map's size, making it more adaptable to various exhibition layouts. The modular design also adds flexibility, allowing curators to adjust the map to fit the specific needs of each exhibition. This solved a major issue with the previous prototype, which was more rigid and less suited for different gallery setups.

Although this phase of the project has resolved many of the challenges from the earlier version, some issues might still arise in the next stage. Overall, we consider this phase to be a success. We effectively addressed the main limitations of the original design and delivered a more versatile and functional solution.

Conclusion

The successful completion of the Phase 1 Audio-Tactile Map (ATM) project for the Gannett Art Gallery represents a significant step toward improving accessibility for visually impaired individuals. Through meticulous design, our team developed a modular, scalable prototype that integrates mechanical and electronic components to create an interactive and user-friendly experience based on the previous prototype. By leveraging advanced tools such as SolidWorks for CAD modeling and ANSYS for stress analysis, we ensured the structural integrity and functionality of the map.

Despite challenges related to the precision of 3D printing and integration of electronic components, our iterative approach allowed us to overcome many of these

limitations. The modular design enhances adaptability, making it possible for the map to accommodate various exhibitions with minimal adjustments. Moving forward, the insights gained in this phase will guide further refinements in both design and functionality to ensure the ATM continues to meet the evolving needs of the Gannett Art Gallery and its visitors.

In summary, this project not only addressed the initial limitations of the prototype but also laid the foundation for future enhancements, ultimately contributing to a more inclusive and accessible gallery experience.

Appendix

Calculate the pressure:

$$P = \frac{F}{A} = \frac{0.4375 \text{ lb}}{256 \text{ in}^2} = 1.71 \times 10^{-3} \text{ psi}$$

Calculate the deflection at center by following the formula,

$$y_{max} = \frac{0.142pa^4}{Et^3[2.21(\frac{a}{b})^3+1]} = \frac{0.142x(1.71 \times 10^{-3} \text{ psi})x(16 \text{ in})^4}{(340 \times 10^3 \text{ psi})x(0.5 \text{ in})^3[2.21(\frac{16 \text{ in}}{16 \text{ in}})^3+1]} = 0.1166 \times 10^{-3} \text{ in}$$

Compute the bending stresses at the plate center using formula,

$$\sigma = \frac{0.75pa^2}{t^2[1.61(\frac{a}{b})^3+1]} = \frac{0.75x(1.71 \times 10^{-3} \text{ psi})x(16 \text{ in})^2}{(0.5 \text{ in})^2[1.61(\frac{16 \text{ in}}{16 \text{ in}})^3+1]} = 0.503 \text{ psi}$$

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